



Final Year Design Project Report

PROJECT NAME

BS - Computer Science

Submitted by:

Student Name: **Reg. ID:**

Student Name **xxxxxx**

Student Name **xxxxxx**

Student Name **xxxxxx**

Project Supervisor:

Supervisor Name

[Designation]

Month, YEAR [2025]

Faculty of Engineering, Sciences and Technology

Preface

Preface is an introduction of your report. What is it about? What does it signify in general, and what kind of knowledge does it hold in general. By reading at the preface one should be able to understand the concept of this report. You may choose to write in 1st person singular or plural language i.e. you may use I or We in this section. You may also share your overall experience of FYP in this section. You must select this paragraph and write your own preface.

Generative AI and Originality Declaration

We, the following students, solemnly declare that the work presented in our undergraduate project, titled **“Project Name”**, is solely our work with no significant contribution from any other person or Generative AI Tools. Small contribution/help wherever taken has been duly acknowledged/cited and that complete project has been written by us in accordance with the latest plagiarism policy declared by HEC and our respective university in-line with the policy for use of Generative AI Tools.

We understand the zero-tolerance policy of the HEC and IQRA University towards plagiarism. Therefore, we as Authors of the above titled project declare that no portion of our project has been plagiarized and any material used as reference is properly referred/cited.

We undertake that if we are found guilty of any plagiarism in the above titled project even after award of degree, the University reserves the right to withdraw/revoke our degrees and that HEC and the University have the right to publish our names on the HEC/University Website on which names of students are placed who submitted plagiarized projects.

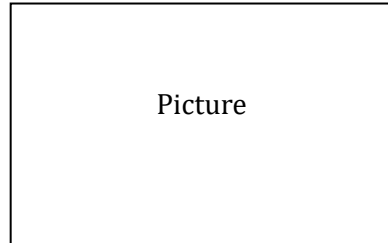
Student Names	IDs	Signatures

Acknowledgements

This page should consist of the acknowledgements to the people, companies and institutions that have been helpful to the author in compiling the reports. It is normal practice to thank the Head of Institute for the use of facilities with which the project was carried out, the head of department, the supervisor for his/her suggestions and guidance and any other member of the academic and technical support staff who have made a significant contribution to the success of the project, and finally your family member (that is optional).

Introduction to Group Members

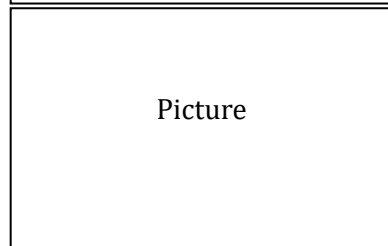
Students must paste their passport size picture instead of cut paste from the groups or casual photograph. Moreover DELETE this sentence too.



Member 1 Name (GL) (Student ID)

System Programmer / UI-UX Creative Designer / Contact

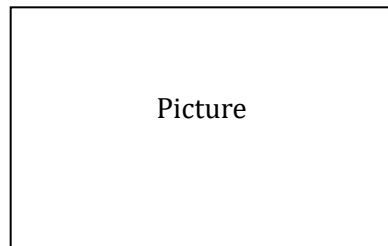
No:0300-XXXXXXX Email:abc@iqra.edu.pk



Member 2 Name (Student ID)

System Quality Analyzer / 3D Model Designer Contact

No:0300-XXXXXXX Email:abc@iqra.edu.pk

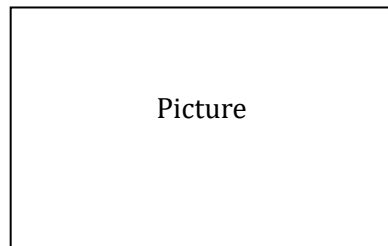


Member 3 Name (Student ID)

Developer / 3D Animator Contact

No:0300-XXXXXXX

Email:abc@iqra.edu.pk



Member 4 Name (Student ID)

Concept Designer / 3D Model Designer Contact

No:0300-XXXXXXX Email:abc@iqra.edu.pk

CERTIFICATE OF COMPLETION

This is to certify that the following students

Group Member 1 Name	Student ID
Group Member 2 Name	Student ID
Group Member 3 Name	Student ID
Group Member 4 Name	Student ID

Have successfully completed their Final Year Project named

Project Title

In the partial fulfillment of the Degree of Bachelor of Science in Computer Science

Supervisor Name

Designation

Abstract

The abstract is a brief summary of your research. Typically, an abstract should be one page and/or less than 350 words. Your abstract should consist of concise and precise information to inform the reader of the content of the report, what the project was about, the main aim of the project, how the work was undertaken and major conclusions drawn from the work performed. It is important not to confuse an abstract with an introduction. The first sentence should give the subject of the report and the last sentence should state the primary conclusion of the report. The abstract should be written in the present tense. It should not include illustrative material such as formulas, diagrams, and charts. The abstract page is numbered ii. It should conclude with a short entry 'Keywords', nominating several keywords by which a computerized library would find the project.

Keywords: Write your own

SDGs Feedback by Students

SDG Name	Target	Feedback by Student
No Poverty (SDG 1)	This project provides affordable access or services that improve livelihood opportunities for underprivileged communities.	The project helps users build self-reliance and reduce poverty by offering sustainable and accessible support solutions.
Zero Hunger (SDG 2)	This project supports improved food access, nutrition tracking, or communication of dietary needs.	Users are able to better communicate health or nutrition concerns, helping reduce hunger-related risks.
Good Health and Well-being (SDG 3)	Through its implementation, this project helps users enhance their mental and physical health while decreasing feelings of isolation or stress.	User's mental health improves. Users develop greater self-assurance in their social and work environments because of this treatment solution.
Quality Education (SDG 4)	This project enables educational settings to include students who might need special assistance so these students can fully participate in their studies.	Through its use, teachers attain improved knowledge about students needing support, thus enhancing their capabilities in inclusive learning.

Gender Equality (SDG 5)	This project ensures equal access for all genders and addresses gender-specific needs within its functionality.	The project empowers users of all genders, especially those who may face systemic barriers to access.
Clean Water and Sanitation (SDG 6)	This project allows users to report water or sanitation issues, improving health outcomes in communities.	The tool supports better hygiene communication and awareness in both personal and institutional environments.
Affordable and Clean Energy (SDG 7)	This project is designed to operate efficiently with minimal energy use, making it viable in low-resource settings.	Students acknowledged its energy efficiency, allowing continuous service even in areas with limited power.
Decent Work and Economic Growth (SDG 8)	This project enhances users' ability to access job opportunities and communicate effectively in professional settings.	Students observed better workplace confidence, increasing employment potential for users.

Industry, Innovation and Infrastructure (SDG 9)	This project leverages innovative technology to improve accessibility and solve real-world issues.	Students viewed the project as a forward-thinking solution that contributes to equitable technological development.
Reduced Inequalities (SDG 10)	This project supports marginalized communities by offering tools that address social and economic barriers.	The specially designed features of this project aim to eliminate inequalities by serving an underrepresented user group.
Sustainable Cities and Communities (SDG 11)	This project encourages inclusive participation in city life, public services, and local engagement.	Students said it helps foster more connected and accessible communities for all citizens.
Responsible Consumption and Production (SDG 12)	This project promotes sustainable use of resources and reduces dependence on physical alternatives.	The project reduces waste and encourages digital solutions over traditional resource-intensive methods.
Climate Action (SDG 13)	This project spreads awareness and supports communication around environmental and climate-related actions.	Students believe it empowers more people to engage in sustainability initiatives and environmental conversations.

Life Below Water (SDG 14)	This project may be used in marine conservation campaigns to educate and involve diverse communities.	The project enables participation in ocean-related awareness and inclusive advocacy.
Life on Land (SDG 15)	This project supports initiatives that promote biodiversity, land preservation, and environmental education.	Students appreciated its potential to involve more people in conservation and eco-friendly actions.
Peace, Justice and Strong Institutions (SDG 16)	This project enables users to report issues, seek support, and engage with institutional frameworks.	Students said it strengthens transparency and user interaction with public systems and justice services.
Partnerships for the Goals (SDG 17)	This project is built through collaboration across sectors like education, tech, and health.	Students recognized how partnerships enhanced the project's overall effectiveness and reach.

CCP Feedback by Students

CCP Attribute	WP1 and some or all WP2 to WP7:	FYDP to CCP Mapping How it is addressed in FYDP
Depth of knowledge required	WP1: one or more of ACM Knowledge Ara	
Range of conflicting requirements	WP2: wide-ranging or Conflicting technical, engineering/computing and other issues	
Depth of analysis required	WP3: no obvious solution	
Familiarity of issues	WP4: Involve infrequently encountered issues	
Extent of applicable codes	WP5: outside problems encompassed by standards and codes of practice	
Extent of stake-holder involvement and conflicting requirements	WP6: diverse groups of stakeholders with widely varying needs	
Interdependence	WP7: many component parts or sub-problems	

Knowledge Areas in ACM

Sr.#	ACM Knowledge Area	Your Project Mapping to ACM Knowledge Area
1	AL-Algorithms and Complexity	
2	AR-Architecture and Organization	
3	CN-Computational Science	
4	DS-Discrete Structures	
5	GV-Graphics and Visualization	
6	HCI-Human-Computer Interaction	
7	IAS-Information Assurance & Security	
8	IM-Information Management	
9	IS-Intelligent Systems	
10	NC-Networking and Communication	
11	OS-Operating Systems	
12	PBD-Platform-based Development	
13	PD-Parallel and Distributed Computing	

14	PL-Programming Languages	
15	SDF-Software Development Fundamentals	
16	SE-Software Engineering	
17	SF-Systems Fundamentals	
18	SP-Social Issues and Professional Practice	

PBL Feedback by Student

Feedback by Student
The following problems were solved: In our project, Problem-Based Learning (PBL) helps in identifying real-world challenges in stuttering detection. It guides model improvements by analyzing speech variations, optimizing deep learning algorithms, and enhancing the Gradio-based user interface for better usability.

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CHAPTER # 1

Introduction

1. INTRODUCTION

1.1 BACKGROUND

This should introduce the reader to the subject described in the report, **what the project was about and why it was undertaken**. It should be written in a manner sufficient to allow a reader unfamiliar with the project to appreciate any background knowledge necessary to understand the report.

Describe the function, show the **block/system** diagram, and give the performance specifications as they appear in your final proposal. Show that you know what variables are important in your project's performance, and what values they should take on. Describe briefly the subprojects into which the project has been divided. Introduce the structure of the report (brief summary about what you will cover in which chapter in separate paragraphs).

1.2 PROBLEM STATEMENT

The aim of the project should be clearly stated with sufficient explanation to make these easily understandable. Write succinct definition of the problem. Defines and limits the 'Scope' of the effort. Clearly describe how you tackled the problem. Preferred to explain each problem in separate subheading. It is preferred to mention problem regarding different stakeholders, and technology related issues. Provide enough information for knowledgeable person where difficulty or lack of time has caused a change in the aims of the project. The nature of any changes in aims and the reasons for these changes should be explicitly stated within this Chapter.

1.3 Introduction:

1.4 Objectives:

1.5 Project Scope:

1.6 Outcomes of Proposed System:

CHAPTER # 2

Literature Review

2. Literature Review

2.1 Research Paper Summaries

2.2 Researches in Pakistan

2.3 Literature Analysis:

CHAPTER # 3

Methodology

3. METHODOLOGY

3.1 Implementation:

3.2 Methodology:

3.3 Working:

3.4 Functional Requirements:

3.5 Non-Functional Requirements:

CHAPTER # 4

Implementation and Testing

4.1 Summary

39

CHAPTER # 05

Experiments and Results

5.1 Hardware Requirement

5.2 Software Requirement

5.3 Results

CHAPTER # 6

Project Costing

6.1 Summary

CHAPTER # 7

Conclusion and Future Work

7.1 Summary

REFERENCES

REFERENCES

At the end of your work, list full details of all of the sources which you have cited in your text in a section headed *References*, in numeric order. References listed must follow IEEE formatting guidelines (see reference examples overleaf). Your reference list should allow anyone reading your work to identify and find the material to which you have referred.

In IEEE style your reference list should be formatted in the following way:

- Align references left
- Single-space each entry, double-space between every new entry
- Place number of entry at left margin, enclose in square brackets [] Indent text of entries

Citations/references with multiple authors

If you choose to mention the author(s) of a source whilst citing it in the text of your work, if there are three or more you can abbreviate them using ‘et al.’ e.g. During their research, Fan, et al. [4] discuss lasers in detail. However, in general you do not need to mention the authors by name, just use the numeric citation in square brackets. In your full reference list at the end however, you always give the authors’ names. In the reference list you can only abbreviate these using ‘et al.’ if there are six or more authors.

Reference examples

There are standard reference formats for most types of document. Below are examples of the most common types of document you might want to reference. Each of the following gives a suggested standard format for the reference followed by examples for the different document types.

Book

[Ref number] Author’s initials. Author’s Surname, *Book Title*, edition (if not first). Place of publication: Publisher, Year.

- [1] I.A. Glover and P.M. Grant, *Digital Communications*, 3rd ed. Harlow: Prentice Hall, 2009.

Book chapter

- [Ref number] Author's initials. Author's Surname, "Title of chapter in book," in *Book Title*, edition (if not first), Editor's initials. Editor's Surname, Ed. Place of publication: Publisher, Year, page numbers.
- [2] C. W. Li and G. J. Wang, "MEMS manufacturing techniques for tissue scaffolding devices," in *Mems for Biomedical Applications*, S. Bhansali and A. Vasudev, Eds. Cambridge: Woodhead, 2012, pp. 192-217.

Electronic Book

- [Ref number] Author's initials. Author's Surname. (Year, Month Day). *Book Title* (edition) [Type of medium]. Available: URL
- [3] W. Zeng, H. Yu, C. Lin. (2013, Dec 19). *Multimedia Security Technologies for Digital Rights Management* [Online]. Available: <http://goo.gl/xQ6doi>

Note: If the e-book is a direct equivalent of a print book e.g. in PDF format, you can reference it as a normal print book.

Journal article

- [Ref number] Author's initials. Author's Surname, "Title of article," *Title of journal abbreviated in Italics*, vol. number, issue number, page numbers, Abbreviated Month Year.
- [4] F. Yan, Y. Gu, Y. Wang, C. M. Wang, X. Y. Hu, H. X. Peng, et al., "Study on the interaction mechanism between laser and rock during perforation," *Optics and Laser Technology*, vol. 54, pp. 303-308, Dec 2013.

Note: the above example article is from a journal which does not use issue numbers, so they are not included in the reference.

E-Journal article

PDF versions of journal articles are direct copies of the print edition, so you can cite them as print journals.

[Ref number] Author's initials. Author's Surname. (Year, Month). "Title of article." *Journal Title* [type of medium]. volume number, issue number, page numbers if given. Available: URL

- [5] M. Semilof. (1996, July). "Driving commerce to the web-corporate intranets and the internet: lines blur". *Communication Week* [Online]. vol. 6, issue 19. Available: <http://www.techweb.com/se/directlinkcgi?CWK19960715S0005>

When you are compiling your reference list you may abbreviate journal titles:

For a list of IEEE abbreviations go to:

https://www.ieee.org/documents/trans_journal_names.pdf

For non IEEE journal abbreviations go to:

<http://www.bath.ac.uk/library/help/infoguides/abbreviations.html>

For further information on the common abbreviations of words used in references for the IEEE style go to:

http://www.ieee.org/documents/style_manual.pdf

Conference papers

[Ref number] Author's initials. Author's Surname, "Title of paper," in *Name of Conference*, Location, Year, pp. xxx.

- [6] S. Adachi, T. Horio, T. Suzuki. "Intense vacuum-ultraviolet single-order harmonic pulse by a deep-ultraviolet driving laser," in *Conf. Lasers and Electro-Optics*, San Jose, CA, 2012, pp.2118-2120.

Standard abbreviations may be applied to the title of the conference. For a table of abbreviations go to: <http://www.ieee.org/documents/ieeecitationref.pdf>

Reports

The general form for citing technical reports is to place the name and location of the company or institution after the author and title and to give the report number and date at the end of the reference. If the report has a volume number add it after the year.

[Ref number] Author's initials. Author's Surname, "Title of report," Abbreviated Name of Company., City of Company., State, Report number, year.

[7] P. Diamant and W. L. Luptakin, "V-line surface-wave radiation and scanning," Dept. Elect. Eng., Colombia Univ., New York, Sci Rep. 85, 1991.

Patents

[Ref number] Author's initials. Author's Surname, "Title of patent," Country where patent is registered. Patent number, Abbrev of Month Day Year.

[8] J. P. Wilkinson, "Nonlinear resonant circuit devices," U.S. Patent 3 624 125, July 16 1990.

Note: Use "issued date" if several dates are given.

Standards

[Reference number] *Title of Standard*, Standard number, date.

[9] *Shunt power capacitors*, IEEE standard 18-2012, 2013.

Theses/Dissertations

[Ref number] Author's initials. Author's Surname, "Title of thesis," Designation type, Abbrev. Dept., Abbrev. Univ., City of Univ., State, Year.

[10] J. O. Williams, “Narrow-band analyser,” Ph.D. dissertation, Dept. Elect. Eng., Harvard Univ., Cambridge, MA, 1993.

Datasheets

[Ref number] Author’s initials. Authors Surname, “Title of Datasheet,” Part datasheet, Publication date [Latest revision date].

[11] Texas Instruments, “High speed CMOS logic analog multiplexers/demultiplexers,” 74HC4051 datasheet, Nov. 1997 [Revised Sept. 2002].

Online Documents

If you are using documents such as a report, conference paper, standard, patent or thesis online and it also exists as an identical print equivalent i.e. with the same format and pagination, it can be usually be referenced as the print version.

If it is e-only, you can make the standard reference template an electronic version by adding the material type in square brackets

e.g. [Online] after the document title. If there is no specific document title you can place this after the document number (e.g. patent number).

At the end of the reference add: Available: URL. See below for an example of an online patent:

[12] M.R. Brooks, “Musical toothbrush with adjustable neck and mirror,” U.S Patent 326189 [Online], May 19 1992. Available: <http://goo.gl/VU1WEk>

Websites

Note: Include as much of the key information as you can find for a given website. If a web page has no personal author, you can use a corporate author. Failing that, you can use either Anon. (for anonymous) or it is permissible to use the title of the site.

[Ref number] Author's initials. Authors Surname. (Year, Month. Day). *Title of web page* [Online]. Available: URL

[13] BBC News. (2013, Nov. 11). *Microwave signals turned into electrical power* [Online]. Available: <http://www.bbc.co.uk/news/technology-24897584>

[14] M. Holland. (2002). *Guide to citing internet sources* [Online]. Available: http://www.bournemouth.ac.uk/library/using/guide_to_citing_internet_source.html

ANNEXURES

Fig 2.1 Use Case Diagram:

Fig 2.2 Context Diagram:

Fig 2.3 Workflow Diagram:

Fig 2.4 Sequence Diagram:

Fig 2.5 ERD Diagram:

Fig 2.6 Gantt Chart FYP I:

Fig 2.7 Gantt Chart FYP II:

Code:

General Guidelines

- Begin each chapter on a new page.
- Each chapter should have a small introduction at the beginning of the chapter. Introduction must link to the previous chapter. It is one or more than one paragraph but not more than one page that introduces the reader to the subject. The introduction presents basic background material, the history of the problem and contains the key sentence outlining the subjects to be discussed.

The total report length should be **80- 120 pages**; most projects are somewhat shorter. There is no value in trying to artificially lengthen your project by ‘padding’ it. Each project is unique and has its own natural length, and you will probably know when you have said everything that needs to be said.

Typing and size of paper

- I. The report is to be typewritten on one side of the paper on international size A4 paper (297mmX210mm). This paper must be good quality bond (70-90 gsm).
- ii. Reports length should be 100-120 pages.
- iii. Use Times New Roman, size 12 font throughout the reports.

Use 1.5 or double spacing.

Page number and Chapter number

Use lower case Roman numerals for preliminary pages

- I. Title page (not numbered on page)
- ii. Abstract

Table of Contents

The text of the report begins with Roman number 1. Number all pages. Page numbers can be inserted either at the bottom/top right or the bottom/top center.

All appendices should number as A-1, A-2, etc. for pages under appendix A, and B-1, B-2, etc. for pages under appendix B (See Table of Contents.).

A hierarchical numbering scheme for chapter numbering shall be used. For instance, use 1 for chapter one, 2 for chapter 2, 1.1 for the subsection 1 of chapter 1, etc. (See the Table of Contents).

Margin boundaries

- I. 1 -inch left margin.
- ii. 0.5-inch margin on the other three sides.

Diagrams and figures

Figures and tables should be inserted in the text in one of the three places.

A full page figure or illustration must be inserted on the left hand side facing the typescript which described it.

Small figure should be incorporated in the text with the legend appearing below (not recommended).

Each graph, figure, etc., should have a figure number and title typed below it. The type style should be the same as the text. Figures should be numbered by chapters (Fig. 1.1, Fig 1.2, Fig. 2.1, etc). explain each figure by referring to its number (e.g. in Fig 1.1), don't assume any figure is self-explanatory. Whichever numbering system you use, make sure that you follow the same system for tables and equations, also explain then as figures.

Line drawings, graphs, and monograms should be in bold clear lines. Where graphs, diagrams and figures cannot be mounted vertically on the page these are to be mounted and labeled in such a way that they can be read from the right hand side(90⁰ on the page) of the page .

All the axes of graphs are to be labeled with the parameter and its units. Information on illustrations and graphs such as labels, scales etc. must be typewritten.

Photocopying

All the figures, etc. must be reproduced by an electronic or electrostatic or photographic method which is known not to fade.

Fixing of photograph

Full page photographs should be bound into the report. Small photographs must be firmly fixed to the paper. An alternative is to use color photocopying or digital processing.

Tables

Each table should be numbered consecutively (Table 1, Table 2) or by chapter (Table 1.1, Table 1.2, Table 2.1). Table number should be centre above the top of the table and be followed on the next line by a brief descriptive caption, preferably in cap. The type should be the same as the text. Refer to each table in text by number “In Table 1, one can clearly see.....” The same rules for location of figures apply to tables.

TABLE 1. MEASURED RESISTOR VALUES AND THE METER ERROR

Nominal Value Marked	Measured Value	Error (%)

Equations

Centre each equation on a separate line. Number equations consecutively in parentheses at the right margin. Equations may be referenced by number in the text, using parentheses around the number.

$$Y(t) = \int \sin(x) dx \quad (1)$$

Units

The S.I. system of units is to be used throughout. Where difficulties are introduced by quotation of imperial units from reference source, these should be accompanied by the appropriate conversion to S.I. units in parentheses.